

Queueing-Theoretic Performance Model for Continuous-Batching LLM Inference

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Motivation. Production LLM serving systems (vLLM, SGLang, TensorRT-LLM) combine continuous batching with paged KV-cache management to maximize GPU utilization under bursty workloads. Operators face a dimensioning problem: provision KV-cache capacity and batch width to meet SLOs on TTFT (time-to-first-token) and TPOT (time-per-output-token), but currently rely on expensive empirical sweeps rather than analytical tools. This project develops a queueing-theoretic model of such a system and uses it to derive optimal admission policies.

Approach. We model the serving engine as a processor-sharing queue with state-dependent service rate (per-iteration latency grows sub-linearly with active batch size) and a finite memory pool (KV cache) that caps concurrency. Each request carries a bivariate service demand: prompt length (prefill, compute-bound) and generation length (decode, memory-bandwidth-bound), drawn from empirical distributions (ShareGPT). Arrivals are Poisson in the baseline, with a heavy-tailed extension. The sub-linear service-rate assumption is workload-dependent: at very small batches the system is compute-bound rather than memory-bound, so we expect to fit a piecewise rate function from microbenchmarks rather than rely on a single curve. Three analytical contributions:

1. A two-stage queueing approximation (admission buffer $\rightarrow$ prefill/decode PS pool) that gives a closed-form estimate of mean TTFT as a function of $(\lambda, \text{KV budget } K, \text{max batch } B)$.
2. A Markov chain over (queue length, active batch size, aggregate KV occupancy) solved numerically for tail metrics (p95/p99 TTFT, blocking probability). The state space grows quickly with K and B ; we aggregate KV at the page granularity paged attention already imposes, which keeps the chain tractable for the regimes we target.
3. A dynamic-programming formulation of admission control: given predicted p99 TTFT in each state, derive the accept/reject policy that maximizes goodput subject to the SLO. Compared against FCFS and SRPT baselines.

We restrict to single-model, single-GPU deployments. Tensor parallelism, chunked prefill, and speculative decoding are out of scope for the analytical model and will be listed as limitations in the final writeup.

Validation. A discrete-event simulator (SimPy) is first cross-validated against the analytical model, then against real vLLM measurements. Experiments run on Modal (single A10/L4 GPU) with Qwen 2.5-1.5B and a ShareGPT replay trace, sweeping arrival rate and max-batch configurations. Target: predictions within $\pm 20\%$ of measured p50/p99 TTFT.

Team division of work. Javed (analytical track): Markov chain formulation, closed-form approximation, DP admission policy derivation, numerical solver. Adam (empirical track): SimPy simulator, Modal/vLLM setup, measurement sweeps, plotting infrastructure. Both authors contribute to the final writeup and to the cross-validation step where analytical predictions are compared against simulation and measurement.

Deliverables. (1) Analytical solver in Python; (2) discrete-event simulator in SimPy; (3) vLLM validation sweep with prediction-vs-measurement plots; (4) FCFS vs SRPT vs DP-derived admission policy comparison on p99 TTFT and goodput; (5) 8-page writeup.

Timeline (~ 4 weeks).

Week	Milestone
1	Literature review (Orca, vLLM, Sarathi-Serve, DistServe); formalize Markov chain.
2	Analytical solver + SimPy simulator; internal cross-validation.
3	vLLM measurement sweep on Modal; prediction-vs-measurement plots.
4	DP admission policy + FCFS/SRPT comparison; final writeup.

Key references. Yu et al., *Orca: A Distributed Serving System for Transformer-Based Generative Models* (OSDI '22); Kwon et al., *Efficient Memory Management for Large Language Model Serving with PagedAttention* (SOSP '23); Agrawal et al., *Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve* (OSDI '24); Zhong et al., *DistServe: Disaggregating Prefill and Decoding for Goodput-Optimized LLM Serving* (OSDI '24); Harchol-Balter, *Performance Modeling and Design of Computer Systems: Queueing Theory in Action* (Cambridge, 2013).